Muchen Xi

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Academic Appointment

Rice University, Postdoctoral Research Fellow; Supervisor: Dr. Tianjun Sun — June 2026 – Present

Education

Washington University in St. Louis, Ph.D. (Social and Personality Psychology) — May 2026
Certificate: Quantitative Data Analysis
Dissertation: *Dynamic Individualized Latent Trait (DILT) Model: Bridging Nomothetic and Idiographic Psychometrics* [Link]

Washington University in St. Louis, Master of Arts in Psychology — May 2023
Thesis: *Testing individual differences in consistency by combining density distributions and if… then… contingencies* [Link]

University of Minnesota, Twin Cities, Bachelor of Science in Psychology — May 2021
Major: Psychology (Cum Laude, with High Distinctions)
Minors: Statistics and Integrative Neuroscience

Research Interests

- **Quantitative methods**: Bayesian inference; multilevel, structural equation, time-series, and dynamic systems modeling; intensive longitudinal data (ILD) analysis
- **Psychometrics**: item response theory (IRT); computerized adaptive testing (CAT); individualized and context-sensitive assessment
- **Machine learning / artificial intelligence (AI)**: Gaussian processes; natural language processing (NLP); large language models (LLMs)
- **Substantive interests**: personality processes and dynamics; individual differences; personality development; person–environment fit, interactions, and transactions
- **Applied interests**: personnel selection and workplace assessment; the future of work; employee well-being

Publications

Aschenbrenner, A. J., **Xi, M.**, Cohn, J., Simon, D. A., & Jackson, J. J. (2026). Developing a psychological test battery to measure cognition in daily life. *Behavior Research Methods.* https://pubmed.ncbi.nlm.nih.gov/42498901/

Xi, M., & Jackson, J. J. (2025). Behavioral variability as a function of people, situations, and their interaction. *Journal of Personality and Social Psychology.* Advance online publication. https://doi.org/10.1037/pspp0000563

Xi, M., & Jackson, J. J. (2025). Whether personality changes across the life span and why. In Vail, K (Eds.), *Handbook of the Science of Existential Psychology*. Guilford.

Chen, Y., **Xi, M.**, Montgomery, J., Jackson, J., & Garnett, R. (2024). Idiographic personality Gaussian process for psychological assessment. *Advances in Neural Information Processing Systems (NeurIPS)*, 37. https://arxiv.org/abs/2407.04970

Sassenberg, T. A., **Xi, M.** (Co-first author), Zhao, D., Blain, S. D., & DeYoung, C. G. (2023). Intelligence predicts sensory discrimination ability but not implicit reward learning. *Journal of Individual Differences*, 44(3), 173–180. https://doi.org/10.1027/1614-0001/a000390

Blain, S. D., Sassenberg, T. A., **Xi, M.**, Zhao, D., & DeYoung, C. G. (2020). Extraversion but not depression predicts reward sensitivity: Revisiting the measurement of anhedonic phenotypes. *Journal of Personality and Social Psychology*. https://doi.org/10.1037/pspp0000371

Xi, M., Anderson, & Seak, D. (2019). Effect of extraversion level on performance of the attention task under noise distraction. *The University of Minnesota Undergraduate Journal of Psychology*, 17(3), 10–14. https://sites.google.com/a/umn.edu/sentience/issues

Manuscripts Under Review

Xi, M., & Jackson, J. J. (2026). The Dynamic Contingency Model: Integrating competing theories of personality processes. Manuscript under review at *Journal of Personality and Social Psychology*. Pre-print: https://osf.io/preprints/psyarxiv/nheu4_v1

Xi, M., Erin, E. L., Montgomery, J. M., & Jackson, J. J. (2026). Short, Precise, and Valid: A Computerized Adaptive Approach to Measuring the Big Five. Manuscript under review at *Assessment*.

Simon, D. A., **Xi, M.,** & Jackson, J. J. (2026). The Dynamic Function of Situations in Personality Processes. Manuscript under review at *Journal of Personality and Social Psychology*. Pre-print: https://osf.io/preprints/psyarxiv/zk2je_v3

Liu, S., Xi, M., Yeoh, J., & Jackson, J. J. (2026). LifeSentence: Language models can encode human life-course trajectories from longitudinal panel data. Manuscript under review at *Nature Human Behaviour*. https://arxiv.org/abs/2606.11220

Manuscripts in Preparation

Xi, M., Sun, J., Sun, T., & Jackson, J. J. Multi-Level Examination of Inter-personal behavior, Job performance, and Person-Team Fit. *Manuscript in preparation; planned submission to Personnel Psychology*. Pre-registered: https://osf.io/pm56d

Xi, M., Wang, P., Pu, J., Hoover, J., & Sun, T. Measurement biases in AI-based psychological interviews. *Manuscript in preparation; planned submission to Psychological Science*.

Xi, M., Rauthmann, J., & Jackson, J. J. From theories to models: understanding dynamic individuality. *Manuscript in preparation; planned submission to Psychological Review*.

Xi, M., Montgomery, J. M., Willroth, E., Hill, P., Rauthmann, J., & Jackson, J. J. Dynamic Individualized Latent Trait Model (DILT). *Manuscript in preparation; planned submission to Psychological Methods*.

Xi, M., Simon, D., Ladd-Thomas, D., & Jackson, J.J., Measuring Personality in Context: Situational Changes in Personality States and Item Functioning. *Manuscript in preparation; planned submission to European Journal of Personality*.

Xi, M., Dugan, K., & Jackson, J. J. Planned Missingness Designs for Measuring Personality States in Experience Sampling. *Manuscript in preparation; planned submission to Assessment*.

Xi, M., Sun, J., & Jackson, J. J. Congruence between peer- and self-reported personality manifestations in the workplace. *Manuscript in preparation; planned submission to Organizational Research Methods*. Pre-registration available at: https://osf.io/fvm3c

Xi, M., Beck, E. D., & Jackson, J. J. Understanding behavioral variability by integrating *if…then…* with the density distribution approach. *Manuscript in preparation; planned submission to Journal of Research in Personality*. Pre-print available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4496505

Simon, D., **Xi, M.**, & Jackson, J.J. A dynamic systems approach to modeling personality expression. Pre-registration available at: https://osf.io/wtakc/

Conference Presentations

Xi, M. & Jackson, J. J. (2026). *Dynamic Individualized Latent Trait Model (DILT)*. In **M. Xi** (Chair), *Personality dynamics: From theory to practice* [Symposium]. European Conference on Personality, Edinburgh, UK.

Xi, M., Sun, T., Sun, J., & Jackson, J. J. (2026). Person-Team (PT) fit: Dynamic Processes of Inter-personal Behavior and Job Performance. In A. Sylvara, T. Sun, & I. Oh (Chairs), *The Fate of Our Work Selves: Employment-related Personality Change and Dynamics* [Symposium]. Society for Industrial and Organizational Psychology (SIOP). New Orleans, LA.

Xi, M. & Jackson, J. J. (2026). *Does a Stable Daily Environment Foster Stable Personality Development? Evidence from Longitudinal Panel Data*. In K. Dugan (Chair), *A Million Little Things: Modeling Many Influences on Personality Development* [Symposium]. Society for Personality and Social Psychology. Chicago, IL

Xi, M. & Jackson, J. J. (2025). *Dynamic Transaction of Person, Situation, and Their Interactions.* In **M. Xi** (Chair), *Personality dynamics: From theory to practice* [Symposium]. Biennial Conference of the Association for Research in Personality, Chicago, IL.

Xi, M. & Jackson, J. J. (2025). *Dynamic Individualized Latent Trait Model (DILT).* In G. Liou (Chair), *AI in Selection: Novel Approaches and Applications* [Symposium]. Annual Conference of the Society for Industrial and Organizational Psychology, Denver, CO.

Xi, M., & Jackson, J. J. (2024). *Behavioral Variability as a Function of People, Situations, and Their Interaction*. Single Speaker Presentation of the European Conference on Personality. Berlin, Germany.

Xi, M., & Jackson, J. J. (2024). *Understanding Behavioral Consistency by Integrating If…Then… with the Density Distribution Approach*. In E. Beck (Chair), *Personality dynamics* [Symposium]. Association for Psychological Science. San Francisco, CA.

Jackson, J.J. & **Xi, M.** (2024). *Environmental Consistency Across Life Span.* [Data Blitz] Lifespan Development Summit, Washington University in St. Louis. St. Louis, MO.

Xi, M., & Jackson, J. J. (2024). *Are Consistencies Consistent? Modeling Standard Deviations of Behavior Within and Across Situations*. Poster Presentation Session of the Annual Conference of the Society for Personality and Social Psychology. San Diego, CA.

Xi, M., & Jackson, J. J. (2023). *Variability in Personality States: The Role of Situation* [Data Blitz], Society for Personality and Social Psychology. Atlanta, GA.

Xi, M., Sassenberg, T., Zhao, I.D., Blain, S.D., & DeYoung, C.G. (2020). *Extraversion Predicts Implicit Reward Learning*. Poster Presentation Session of the Annual Conference of the Society for Personality and Social Psychology. New Orleans, LA.

Xi, M., Zhao, I.D., Blain, S.D., & DeYoung, C.G. (2019). *Intelligence, Sensory Discrimination, and Implicit Reward Learning*. Poster Presentation Session of the Annual Conference of the International Society of Intelligence Research. Minneapolis, MN.

Invited Talks

Xi, M. & Jackson, J. J. (2026). *Generalizable Individualization in Modeling Human Behavior* [Invited Talk]. Department of Psychology, Auburn University, Auburn, AL.

Xi, M. & Jackson, J. J. (2025). *Generalizable Individualization in Modeling Human Behavior* [Invited Talk]. Department of Psychology, University of Missouri, Columbia, MO.

Data Collection

Xi, M. & Jackson, J.J. (2025). *Variability in Individuals across Situations, Traits, and Adaptations (VISTA)* (Version 1) [Dataset]. https://osf.io/eu7rz/

Aschenbrenner, A.J., **Xi, M.**, & Jackson, J.J. (ongoing). *Cognitive Variability Battery* [Dataset]

Xi, M., Li, Y., Yu, J., & Pang, J. (ongoing). *Know me Know you* [Dataset]

Wang, P., **Xi, M.**, Sun, T., Xiao, Z. (ongoing). *Multimodality Whole-Persona Dataset* [Dataset]

Technical and Statistical Skills

Programming Skills:

R, Python, Stan, LaTeX, MATLAB, HTML, JavaScript

Analytic Skills:

- Time-series analysis:
  - ARIMA, Vector-Autoregression (VAR) models, Stochastic differential equation models
- Bayesian modeling:
  - Bayesian regression: growth model (e.g., spline & GAM), mixture model, multilevel model
  - Bayesian dynamic models: Dynamic/Continuous-time Structural Equation Modeling (DSEM/ctSEM)
- Machine Learning:
  - Bayesian machine learning (e.g., Gaussian Process, Stochastic Variational Inference)
  - Classical and modern machine learning (e.g., random forests, SVM, neural networks)
  - Dimensionality reduction and clustering (e.g., PCA, t-SNE, UMAP, k-means)
  - Natural language processing (e.g., semantic embedding models, topic modeling, and semantic similarity analysis)
  - Simulation-based validation and model evaluation
- Psychometrics & Measurement
  - Bayesian psychometric models: Factor analysis, Graded-Response Model (GRM), Generalized Partial Credit Model (GPCM)
  - Computerized Adaptive Testing (CAT)
  - Network psychometrics
  - Experience Sampling Method Design (ESM)

Language:

Mandarin (native), English (fluent)

Teaching Experience

Assistant Instructor Washington University in St. Louis

Department of Psychological and Brain Sciences 08/2022-05/2024

Courses:

Psych 353 Psychology of Personality (Fall 2022); Dr. Tammy English

Psych 353 Psychology of Personality (Spring 2023); Dr. Patrick Hill

Psych 353 Psychology of Personality (Spring 2024); Dr. Jessie Sun

Awards & Honors

Auburn University Bridges into Excellence (AUBIE) Fellowship, Auburn University 03/2026
Summer School of Personality Science, European Association for Personality Psychology 08/2024
Certificate in Quantitative Data Analysis, Washington University in St. Louis 05/2024
Mortensen Travel Award, University of Minnesota, Twin Cities 12/2019
Dean's List, University of Minnesota, Twin Cities – All semesters 2017-2021

Service Activities

Washington University in St. Louis
Diversity, Equity, and Inclusion Committee Student Representative 2022-2023
Department of Psychological and Brain Sciences Graduate Students Mentor 2022-2023
Psychology Graduate Student Association, Chair for Organization 2023-2024

Professional Activities

Ad Hoc Reviewer:
Journal of Personality and Social Psychology
Journal of Research in Personality

Memberships

Association for Psychological Science (APS)
Association for Research in Personality (ARP)
European Association for Personality Psychology (EAPP)
Society for Industrial and Organizational Psychology (SIOP)

References

Joshua J. Jackson, Ph.D.
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